

# CoRide: Joint Order Dispatching and Fleet Management for Multi-Scale Ride-Hailing Platforms

Jiarui Jin<sup>1</sup>, Ming Zhou<sup>1</sup>, Weinan Zhang<sup>1</sup>, Minne Li<sup>2</sup>, Zilong Guo<sup>1</sup>, Zhiwei Qin<sup>3</sup>, Yan Jiao<sup>3</sup>, Xiaocheng Tang<sup>3</sup>, Chenxi Wang<sup>4</sup>, Jun Wang<sup>2</sup>, Guobin Wu<sup>4</sup>, Jieping Ye<sup>3</sup>

<sup>1</sup>Shanghai Jiao Tong University, <sup>2</sup>University College London, <sup>3</sup>DiDi AI-Labs, <sup>4</sup>DiDi Research



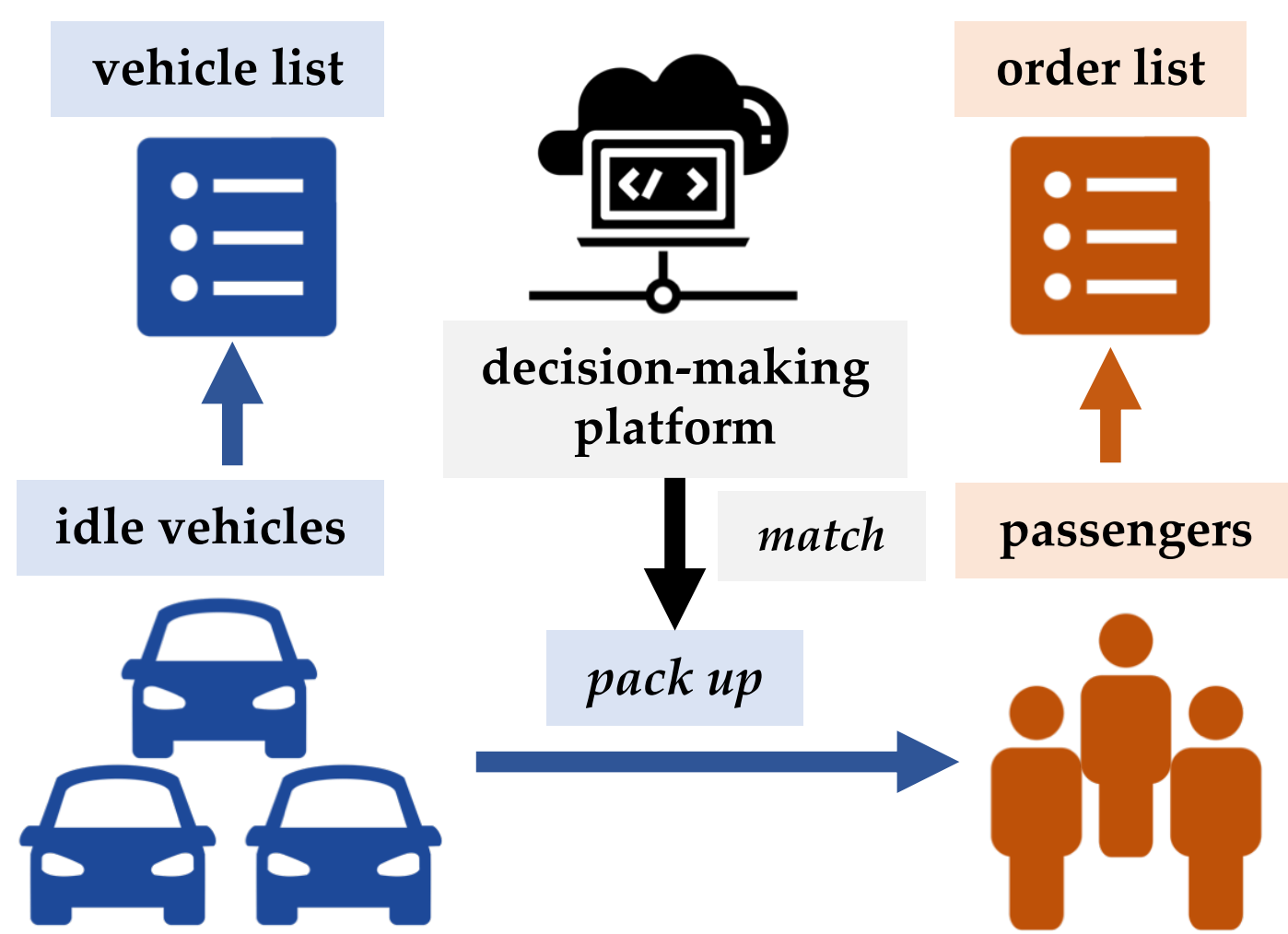
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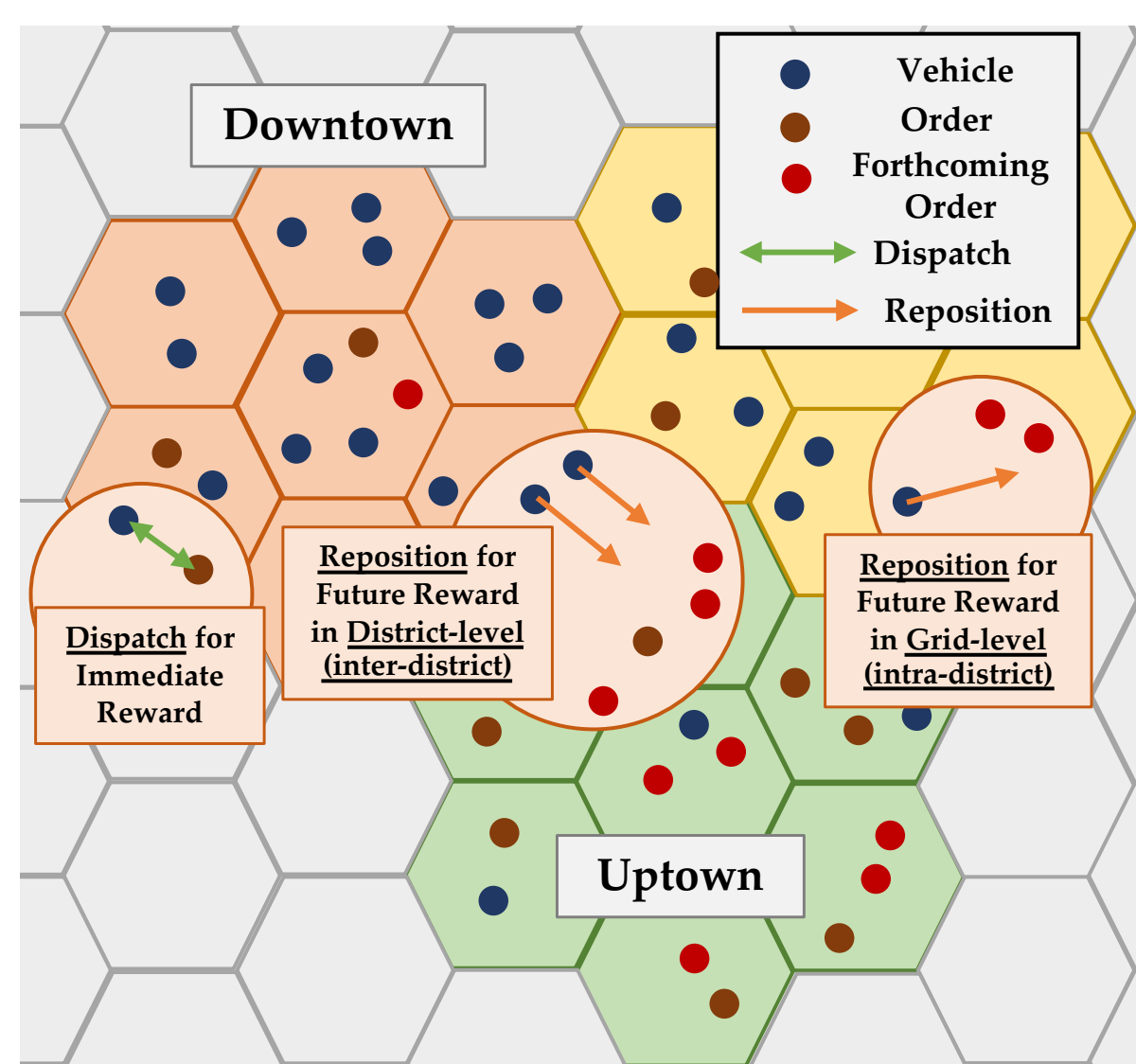
## Background

### Reinforcement Learning for Ride-Hailing Platform

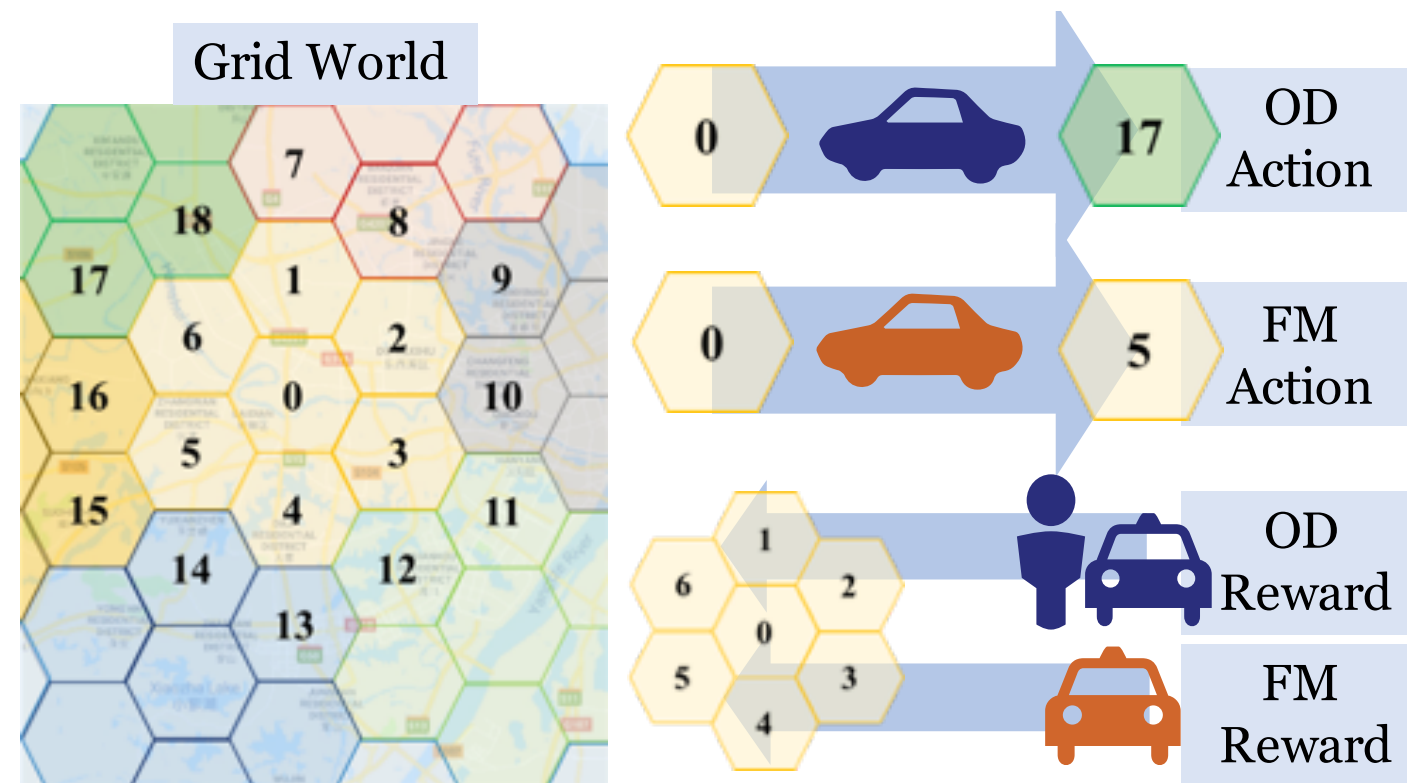


## Task and Preliminaries

- Joint order dispatching and fleet management
  - order dispatching**: match the orders and vehicles in real time to directly deliver the service to the users.
  - fleet management**: reposition the vehicles to certain areas in advance to prepare for the later order dispatching.



- Joint intra-district (grid-level) and inter-district (district-level) allocation
  - intra-district**: *worker* agent -> 1 single grid
  - inter-district**: *manager* agent -> 7 neighboring grids



## Challenges

- Large-scale Agents**: a large number of orders and vehicles -> grid as agent
- Immediate & Future Rewards**: tradeoff between immediate gain and future income -> **different** time credit
- Heterogeneous & Variant Action Space (1<sup>st</sup>)**: joint consider order dispatching and fleet management -> **re-define** action
- Multi-scale Ride-hailing (1<sup>st</sup>)**: conduct ride-hailing in multi-scale way -> **hierarchical** attention mechanism

## Pros of our model

- joint** consideration of different tasks
- stationary** agent number
- different spatial-temporal** credit
- distributed** implementation

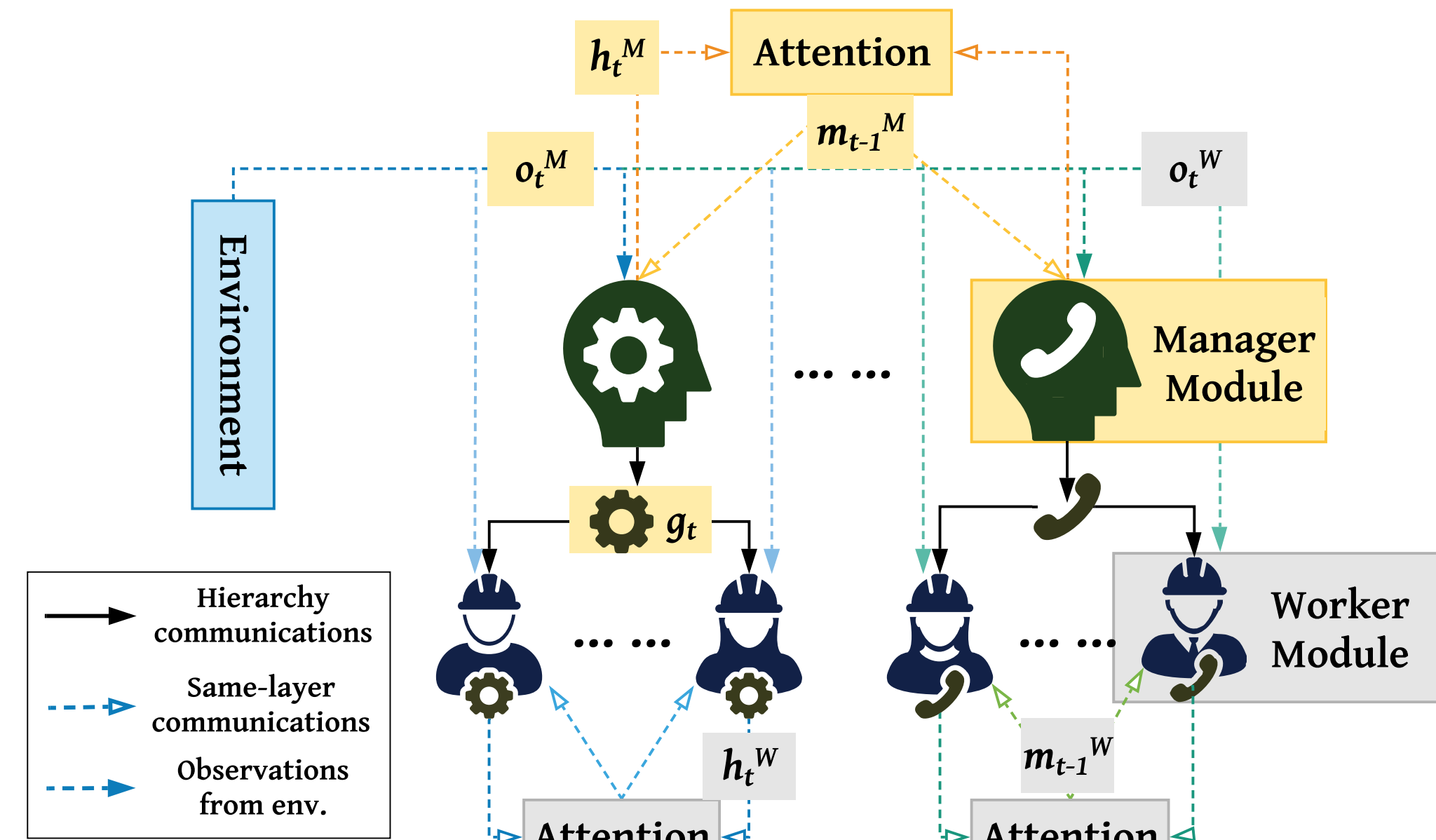
## Experiments

**Metrics**: ADI: Accumulated Driver Income ORR: Order Response Rate  
Performance comparison with respect to the performance of RAN

| City    | City A                |                       | City B                |                       | City C                |                       |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Metrics | Normalized ADI        | Normalized ORR        | Normalized ADI        | Normalized ORR        | Normalized ADI        | Normalized ORR        |
| DQN     | +5.71% ± 0.02%        | +2.67% ± 0.01%        | +6.30% ± 0.01%        | +3.01% ± 0.02%        | +6.11% ± 0.02%        | +3.04% ± 0.01%        |
| MDP     | +7.11% ± 0.05%        | +2.71% ± 0.03%        | +7.89% ± 0.05%        | +3.13% ± 0.04%        | +7.53% ± 0.03%        | +3.19% ± 0.03%        |
| DDQN    | +6.68% ± 0.04%        | +3.19% ± 0.04%        | +7.75% ± 0.06%        | +4.06% ± 0.05%        | +7.62% ± 0.04%        | +4.58% ± 0.05%        |
| MFO     | +6.62% ± 0.03%        | +3.71% ± 0.02%        | +7.91% ± 0.04%        | +4.01% ± 0.02%        | +7.32% ± 0.02%        | +4.60% ± 0.01%        |
| CoRide  | +9.27% ± 0.04%        | +4.23% ± 0.03%        | +8.73% ± 0.03%        | +4.35% ± 0.02%        | +9.06% ± 0.03%        | +4.23% ± 0.04%        |
| CoRide  | <b>+9.80%</b> ± 0.04% | <b>+4.81%</b> ± 0.05% | <b>+8.94%</b> ± 0.06% | <b>+4.89%</b> ± 0.04% | <b>+9.23%</b> ± 0.05% | <b>+5.19%</b> ± 0.04% |

## Our Model (CoRide)

- Action: *manager*: generate **goal**; *worker*: **rank** weight
- Reward: *manager*: **extrinsic** reward; *worker*: **intrinsic** reward



- State:  $S = \langle N_v, N_o, E, N_f, D_o \rangle$ , number of vehicle, number of order, entropy, number of vehicle in FM group, distribution of order features (e.g., price, duration). *manager* and *worker* only differ at scale.

$$N_v \begin{cases} 0 \\ 1 \end{cases} \quad \text{Possible Pairs: } (N_v, N_o) \quad \text{Dispatched Pairs: } (N_v, N_v)$$

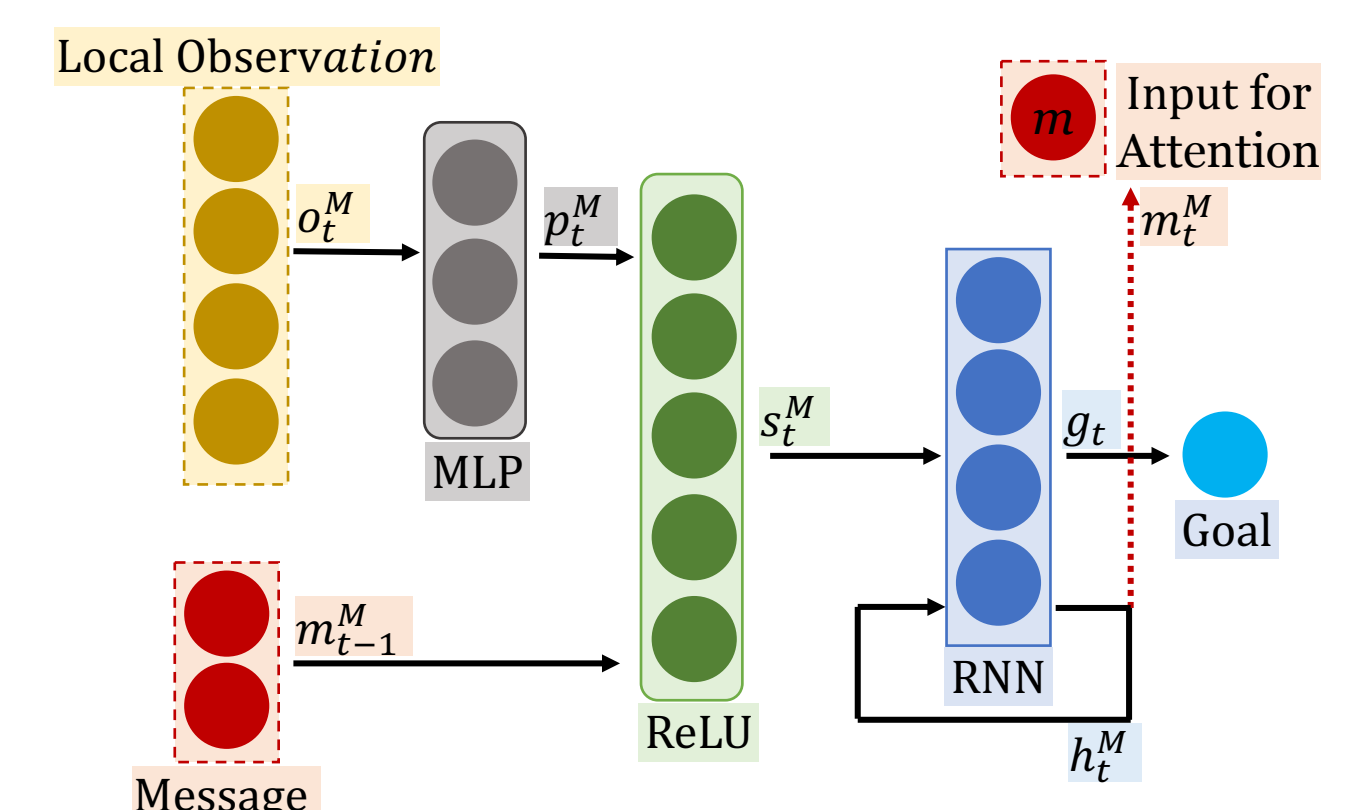
$$E = -k_B \sum_i \rho \ln \rho = -k_B \times \rho \ln \left( \frac{N_v \times N_v}{N_v \times N_o} \right)$$

## Modules

*manager*:

$$r_t^M = r_{ADI} + r_{ORR}$$

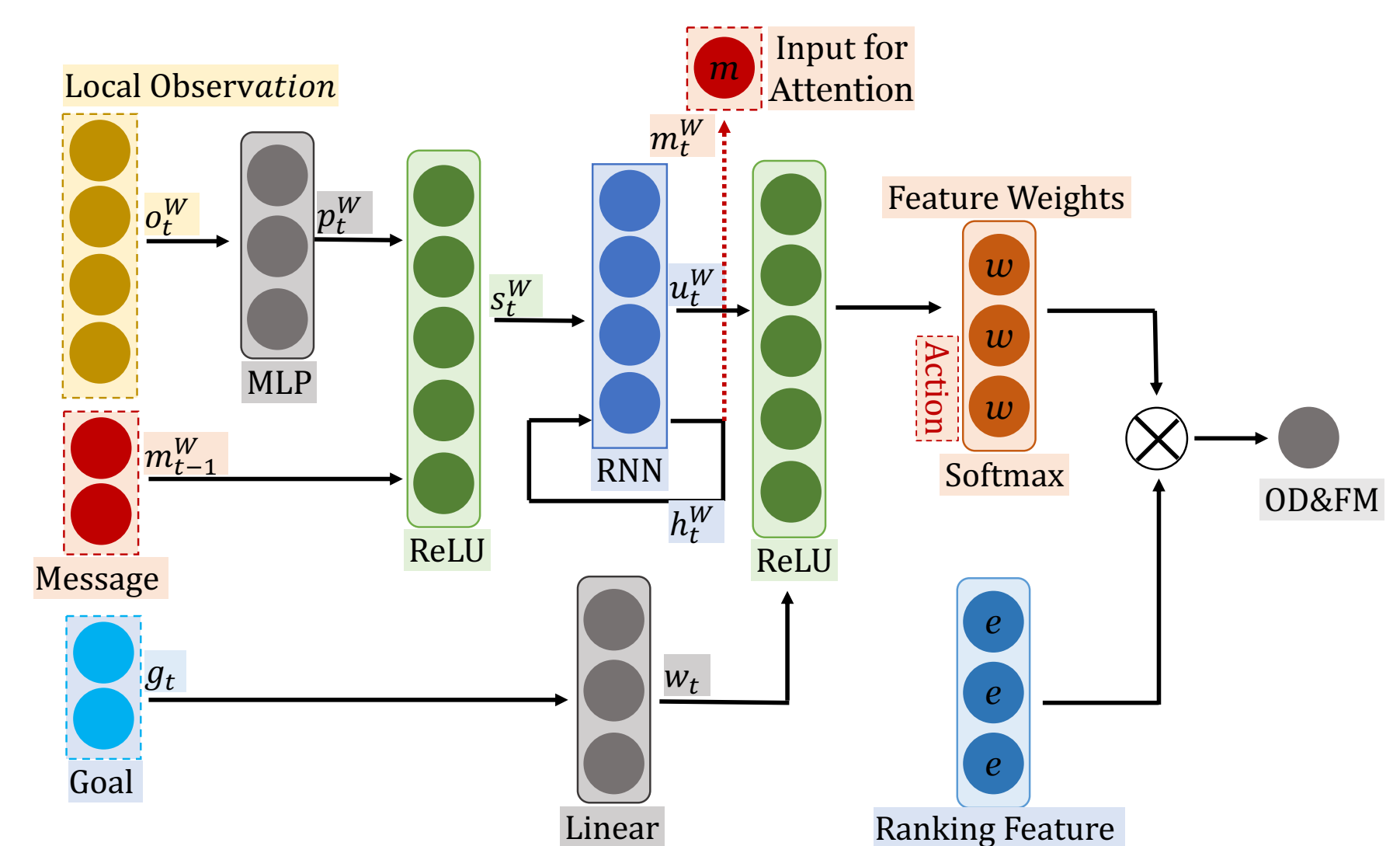
$$r_{ORR} = \sum_{grid} (E - \bar{E})^2 + \sum_{area} D_{KL}(\mathcal{P}_t^o \| \mathcal{P}_t^v)$$



*worker*:

$$r_t^W = \frac{1}{c} \sum_{i=1}^c d_{cos}(o_t^W - o_{t-i}^W, g_{t-i}),$$

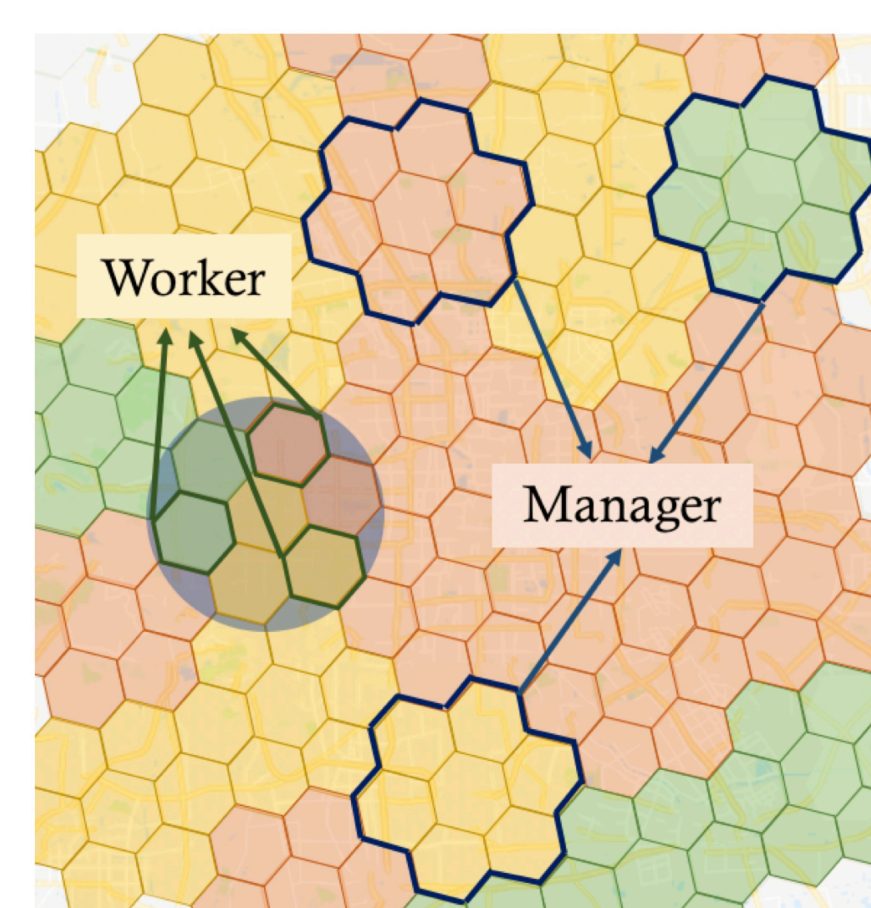
$$d_{cos}(\alpha, \beta) = \alpha^T \beta / (|\alpha| \cdot |\beta|)$$



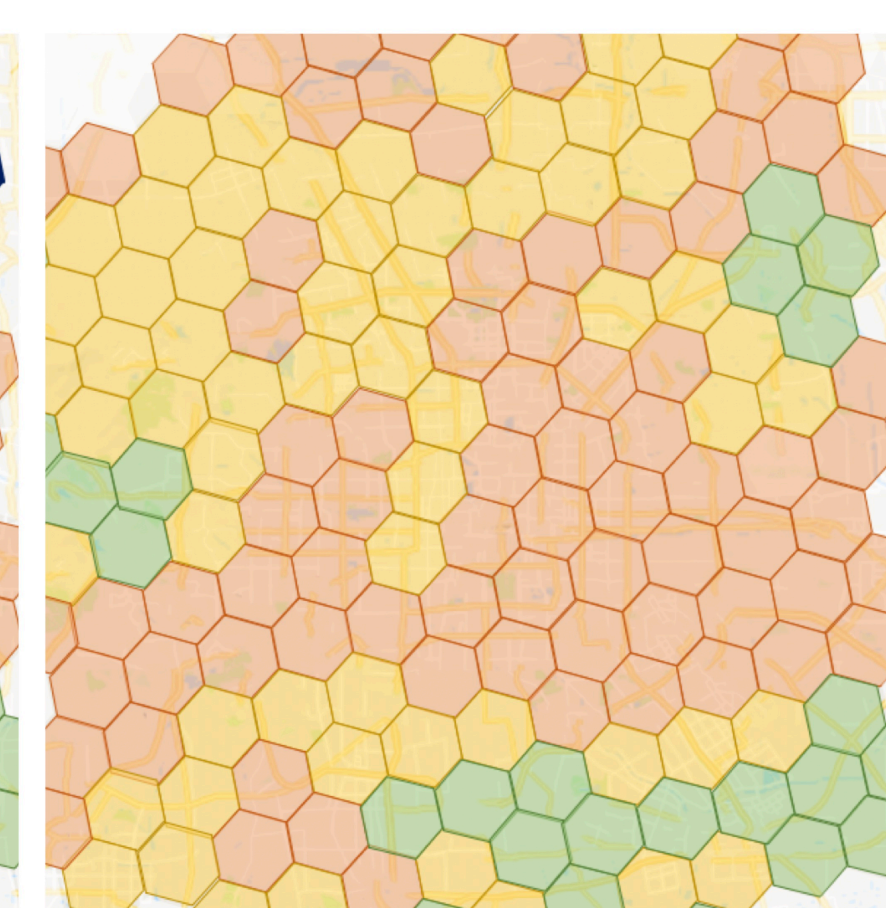
## Algorithm

**Algorithm 1** CoRide for joint multi-scale OD & FM

- Require**: current observations  $o_t^M, o_t^W$ ; mutual communication messages  $m_{t-1}^M, m_{t-1}^W$ .
- for each *manager* in grid world do
  - Generate  $g_t, h_t^M$ .
  - for each *worker* of the *manager* do
  - Generate  $\omega_t, h_t^W$ .
  - Add orders and fleet control items to item space  $\mathcal{I}$ .
  - Rank items in  $\mathcal{I}$ .
  - Generate Selected- $k$  items.
  - end for
  - worker*-level attention mechanism generates  $m_t^W$ .
  - manager* receives extrinsic reward  $r_t^M$ , and its *workers* receive intrinsic reward  $r_t^W$ .
  - end for
  - manager*-level attention mechanism generates  $m_t^M$ .
  - Update parameters via DDPG.



(a) Sampled Attention Value



(b) Demand-supply Gap



Full Paper Link

Jiarui Jin's WeChat

If you have any question, please feel free to contact Jiarui Jin (jinjiarui97@gmail.com)